

Sai Shruti Prakhya

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ABOUT ME

Driven by a lifelong love of learning and problem-solving, I'm passionate about understanding and modeling complex systems through artificial intelligence. My research applies modern machine learning methods, particularly contrastive and multimodal learning, to remote sensing and geospatial data, aiming to build scalable, interpretable AI systems with real-world impact.

EDUCATION

- 2024–Present **Master of Science by Research in Data Science and Artificial Intelligence**, International Institute of Information Technology, Bangalore
Supervisor: Dr. Uttam Kumar
Thesis Topic: Foundation Models for Remote Sensing Data
Courses: Machine Learning, Optimization, Self Supervised Learning, Reinforcement Learning, Few Shot Learning, Visual Recognition, Advanced Reinforcement Learning.
GPA – 3.9/4.0
- 2018–2022 **Bachelor of Technology, Civil Engineering**, National Institute of Technology Karnataka, Surathkal
Minor Degree in Mathematics
Courses: Real Analysis, Complex Analysis, Algebra, Topology, Partial Differential Equations.
CGPA – 8.54/10.00
- 2016–2018 **Higher Secondary Education**, The Valley School, Bangalore
Score – 88.6%

RESEARCH WORK

- Foundation Models for Remote Sensing** April 2025–Present
- Developing **multimodal foundation models** for remote sensing applications including land cover segmentation and classification.
 - Designing modality-specific encoders and cross-attention fusion mechanisms for satellite and SAR data to enable effective multimodal representation learning.
 - Formulating contrastive and reconstructive pretext tasks for scalable, label-efficient self-supervised training across large geospatial datasets.
 - Implementing modular pipelines for fine-tuning and evaluation to ensure reproducibility and efficient deployment across diverse benchmarks.
 - **Tech Stack:** PyTorch, Hugging Face, Swin Transformer, Sentinel-1/2, Google Earth Engine, CUDA
- Discovering Gradient Descent through Deep Reinforcement Learning** March 2025–Present
- Formulating optimizer discovery as a Markov Decision Process (MDP) and developing an RL-based agent to learn adaptive update rules from scratch.
 - Designing reward functions based on optimization convergence for both convex and non-convex landscapes.
 - Experimenting with on-policy and off-policy strategies to enable generalizable, learning-based optimization—towards scalable and autonomous optimizer design.
 - **Tech Stack:** PyTorch, Gymnasium, Deep Q-Learning, Deep MCTS, NumPy

PUBLICATIONS

1. **Spatial Attention-Enhanced Dual-Path Skip Connection Siamese Network for Temporal Land Cover Change Detection using SAR** — Accepted for presentation at *Big Data and AI (BDA)*, 2025. (Deep learning approach for SAR-based change detection)
2. **Agriculture Crop Monitoring for Yield Estimation with Zero-Shot Fruit Detection** — Published as a book chapter in *Remote Sensing of Land Cover and Land Use Changes in South and Southeast Asia, Volume 1* (DOI: [10.1201/9781003396253](https://doi.org/10.1201/9781003396253)). (Vision model for fruit detection and yield estimation)
3. **Predicting Urban Growth Using a Spatio-Temporal Matrix Model: Insights from Bangalore City, India** — Accepted at *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2025. (Spatio-temporal modeling for urban expansion prediction)

WORK EXPERIENCE

Research Assistant, Spatial Computing Lab, IIITB

Dec 2023 – July 2024

- Curated and annotated a real-world horticulture dataset (~110 images) through field visits to government research institutes (IIHR, GKVK); trained and deployed a YOLOv8 model for plant detection (88.8% accuracy) and an SVM for fruit ripeness (99% accuracy); integrated the models into the in-house crop monitoring app *Farmlytics* for yield estimation; co-authored a book chapter on the methodology (**accepted in NASA LULC proceedings**).
- Delivered three hands-on QGIS tutorials on land use and land cover (LULC) classification, reaching 50+ undergraduate and graduate students.
- Drafted a research proposal for satellite-based water quality estimation, integrating spectral indices and supervised learning techniques.
- Designed a task-incremental learning algorithm using Low-Rank Adaptation (LoRA) to mitigate catastrophic forgetting while maintaining performance across sequential vision tasks.

PROJECTS

Multimodal Visual Question Answering with Amazon Berkeley Objects Dataset April–May 2025

- Built a 25k-sample VQA dataset using **LLM-based few-shot** and **chain-of-thought prompting** strategies to generate question–answer pairs.
- Fine-tuned **ViLT** and **BLIP** with **LoRA-based parameter-efficient adaptation**, achieving up to **55% higher accuracy** and a **250× F1 improvement** (BLIP: 0.001→0.2546).
- Engineered scalable pipelines for label adaptation, cosine-similarity answer replacement, and beam search decoding for robust inference.
- **Tech Stack:** Python, PyTorch, Hugging Face Transformers, LoRA, Large Language Models (LLMs)

Feature-Reduced Gumbel-Based Models for Cyber Security Oct–Dec 2024

- Processed **1M+ cyberattack logs** across **10+ threat categories**, performing feature decorrelation and dimensionality reduction (PCA, PPCA, Sparse Autoencoders; ~80% variance retained).
- Modeled rare attack patterns using **Gumbel mixture models** with **Bayesian Gibbs sampling** and Dirichlet priors to identify latent clusters.
- Optimized inference using **JAX-based parallelized sampling**, achieving 3× faster convergence and improved cluster stability.
- **Tech Stack:** JAX, NumPy, PyTorch, Bayesian Modeling, Gibbs Sampling

Image Segmentation and Classification Jan–Mar 2025

- Designed and trained **ResNet-inspired CNN** and **U-Net** architectures for real-time mask detection and segmentation on **6.5k+ images**.
- Achieved **97.2% classification accuracy** and **85% IoU** using **weighted Dice loss** and mixed-precision (**FP16**) training for 2× faster performance.
- Built an automated pipeline for dataset preprocessing, evaluation (Dice, IoU, pixel accuracy), and reproducible experimentation across multiple architectures.
- **Tech Stack:** Python, PyTorch, TensorFlow, scikit-learn, OpenCV, Mixed-Precision Training

POSITIONS OF RESPONSIBILITY

Executive Head – Indian Society for Technical Education (ISTE), NITK Apr 2021 – Apr 2022

- Led a team of **150+ members** across **10 Special Interest Groups (SIGs)**; founded the Social Initiatives Group in collaboration with NITK Spectrum and coordinated with **5+ student committees**.
- Supervised completion of **25+ interdisciplinary projects** and **20+ mentorship programs**, impacting **850+ students**; organized an online project expo and **15+ events** with a collective footfall of **2,500+**.
- Mentored **150+ students** through seminars and mock interviews; enhanced ISTE’s social media presence (**3 handles, 2k+ followers, 100k+ impressions**) with a **15% engagement** and **20% follower** increase.
- Secured sponsorships worth **Rs. 3L+** and successfully managed the flagship technical fest with participation exceeding **2,000**.

Student Community Coordinator – NITK Spectrum Jun 2021 – Jun 2022

- Recruited and led **20+ members** for NITK’s gender and sexuality initiative, increasing social media engagement by **25%**.
- Organized the institute’s first on-campus Mental Health Awareness event and coordinated annual Pride Month celebrations (2021–22) with **200+ participants**.

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- Awarded **Gold Medal** in IEO English Olympiad and **Silver** in Akhila Bharatiya Hindi Olympiad (2015); achieved outstanding performance in ASSET (2014–15) and a **Bronze** in Duke of Edinburgh’s **IAYP** (2014).
- Secured admit to the **Teach For India Fellowship**; volunteered with **Sitara Akka (2021)** to educate underprivileged students.
- Trained in **Bharatanatyam** and **Piano** (6+ years); certified **Beginner Scuba Diver (SSI)**; completed **Yoga training** under Shri B.N.S. Iyengar (Feb 2025).